Exercise Sheet 2

System Safety

This sheet introduces key safety concepts and applies System-Theoretic Process Analysis (STPA) to the CARLA autonomous driving system described in the Course Roadmap. Refer back to that system description for all exercises below.

Hint: You are encouraged to solve these exercises digitally (e.g., using spreadsheets, digital diagrams, or notebooks). This will make it easier to later incorporate your results, diagrams, and evidence into your final safety report.

Terminology

Exercise 2.1: Safety Vocabulary *optional*

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:

1. **Loss:** An adverse consequence. Example: Injury or loss of human life.
2. **Hazard:** A system state or condition that can lead to a loss under certain circumstances. Example: A vehicle moving while a pedestrian is in its path.
3. **Risk:** The combination of the probability of a hazard, the probability of a loss given that hazard, and the severity. Formula:

   Risk ≈ P(hazard) × P(loss | hazard) × Severity

"This exercise establishes a common language for safety so we can distinguish between a bad outcome (Loss), the situation that makes it possible (Hazard), and the mathematical likelihood of it happening (Risk)."

STPA: System-Theoretic Process Analysis

STPA Step 1: Define Purpose and Identify Losses, Hazards, and ODD

Exercise 2.2: ODD Specification

Based *only* on the system description in the course Roadmap (you have not yet seen the training data), define the **Operational Design Domain** (ODD) for the CARLA system. Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed.

For each dimension, state:

1. The **operating conditions** (where the system is considered valid).
2. The **non-operating conditions** (where it is not).
3. How an ODD violation could be **detected at runtime**.

Hint: You are defining the ODD *before* looking at the data. In Exercise Sheet 3, you will compare this ODD to the actual training data and (possibly) discover gaps.

Dimension	Operating Condition (Valid)	Non-Operating Condition	Runtime Detection Method
Weather	Clear, cloudy (no rain/fog).	Heavy rain, fog, snow.	Anomaly/OOD detection scores.
Lighting	Daytime (high sun angle).	Night, low sun (glare).	Average pixel intensity.
Camera	Forward-facing, rigidly fixed.	Loose mount, lens obscuration.	Edge detection/Explainability.
Scene Type	Urban roads, mapped intersections.	Off-road, unmapped highways.	Scene classifier/OOD scores.
Vehicle Speed	<= 50~km/h.	> 50~km/h.	Vehicle speedometer feedback.

"We define the Operational Design Domain to draw a clear boundary around where our model is 'allowed' to work, ensuring we don't expect it to handle conditions like heavy snow or night that it never saw during training."

Exercise 2.3: Losses

STPA begins by identifying the **losses** that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.

For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

"Safety engineering always starts at the end: we identify the absolute worst-case outcomes—like death or property damage—so we can build the entire system specifically to prevent them."

ID	Loss Description	Why Unacceptable
L-1	Injury or loss of human life.	Primary ethical and moral imperative of safety-critical systems.
L-2	Material damage (AV or property).	High financial repair costs and liability for the developer.
L-3	Violation of traffic laws/fines.	Leads to legal liability and potential loss of testing license.
L-4	Loss of public trust/Reputation.	Social and ethical backlash can end the entire AV program.

Exercise 2.4: Hazards

A **hazard** is a system state that, combined with worst-case environmental conditions, can lead to a loss. Hazards are properties of the *system*, not of individual components.

1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

Fill in the following table:

ID	Hazard	Losses	Likelihood	Severity
H-1	Pedestrian in path while vehicle is moving.	L-1, L-2.	Medium	High
H-2	Autonomous mode remains active outside the ODD.	L-1, L-2, L-3.	High	Medium
H-3	Approaching intersection without detecting stop condition.	L-1, L-2, L-3	Medium	High
H-4	Planner commands acceleration when obstacle is in path.	L-1, L-2.	Low	High

"Hazards bridge the gap between normal driving and a crash; they identify the specific system states, like moving while a pedestrian is in the way, that lead directly to our identified losses."

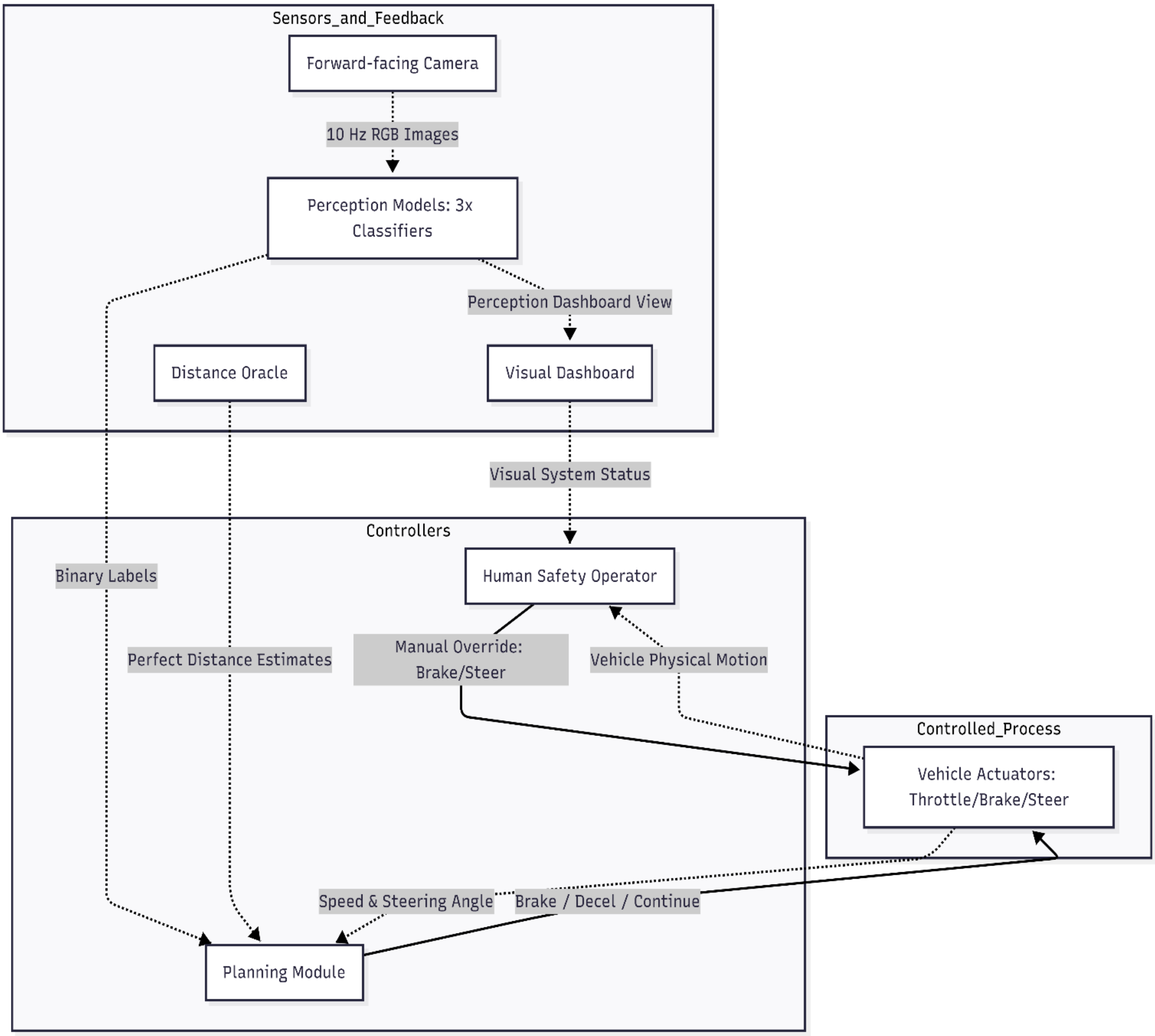
STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:

- All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).
- **Control actions** (solid arrows): commands flowing from controllers to controlled processes (e.g., "brake command").
- **Feedback** (dashed arrows): information flowing back (e.g., "perception labels", "distance estimates", "vehicle speed").

Hint: There are at least two controllers in this system: the planning module and the human operator. The perception models and distance oracle should be considered *sensor/feedback*, not controllers themselves - they provide information to the controllers.



Hint: The traffic-light model only detects presence, not traffic light state. Consider what decisions the planner can (or cannot) safely make with this information.

"Instead of looking at isolated parts, this diagram maps out how 'loops' of control work—showing exactly what commands flow out (solid lines) and what feedback comes back (dashed lines) to keep the vehicle safe.

Control Structure Diagram (Ex 2.5), make sure to point out that the Planning Module is "blind" to raw camera images and only sees the labels, which is why your feedback arrows are dashed.

STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints

Exercise 2.6: Unsafe Control Actions (UCAs)

Using your control structure, identify **unsafe control actions** for at least **two different controllers**. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., "H-1", "H-2").

For each controller, consider all four UCA types:

1. **Not provided** - a required action is missing.
2. **Provided unsafely** - the action is issued when it should not be.
3. **Wrong timing / order** - too early, too late, or out of sequence.
4. **Wrong duration** - applied too long or stopped too early.

Fill in the table below (add rows as needed):

ID	Controller	Control Action	UCA Type	Hazard	Unsafe Scenario
UCA-1	Planner	Continue Driving	Provided unsafely	H-1	Pedestrian detected in path but vehicle does not stop.
UCA-2	Planner	Emergency Brake	Not provided	H-1	Pedestrian within critical distance but brake not issued.
UCA-3	Planner	Emergency Brake	Wrong timing	H-3	Braking issued too late to stop at a detected intersection.
UCA-4	Operator	Manual Override	Not provided	H-2	Automation fails outside ODD; operator fails to take over.
UCA-5	Planner	Emergency Brake	Provided unsafely	H-4	Hard brake issued on highway with no obstacle, causing rear-end.
UCA-6	Planner	Request Decel	Not provided	H-3	Traffic light detected (presence) but no slowdown initiated.
UCA-7	Operator	Manual Override	Wrong timing	H-1	Operator reacts too late (> 1.5s) to a missed ML detection.
UCA-8	Planner	Continue Driving	Wrong timing	H-3	Continues through intersection after light presence was detected.

"Here, we analyze how a controller—either the AI or the human—can mess up by issuing a command too late, not issuing it at all, or issuing it when it shouldn't."

Exercise 2.7: Safety Constraints

For each UCA from Exercise 2.6, derive at least one **safety constraint** - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.

Classify each constraint as either:

- **Model-level** - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
- **System-level** - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

"We turn those 'mess-ups' into rules; if a control action is unsafe, the Safety Constraint is the mandatory requirement we set to ensure that action never happens."

UCA	Causal Scenario	Root Cause	Related Constraint
UCA-1	Detector reports "No Pedestrian".	Out-of-Distribution (OOD) case: pedestrian in wheelchair.	Pedestrian Recall Metric.
UCA-2	Braking not issued.	Overconfident error: low confidence treated as "No detection".	Calibration / ECE Requirement.
UCA-3	Braking issued too late.	High processing delay/latency in the perception pipeline.	Latency threshold requirement.
UCA-4	Operator fails to take over.	"Monitoring fatigue" during the end of a 4-hour shift.	Operator Shift Protocol (4h limit).
UCA-5	False Emergency Brake command.	Spurious correlation: model focuses on the sky/shadows, not objects.	Explainability (GradCAM) overlap.
UCA-6	Planner ignores traffic light.	Flawed internal model: assumes light is green because state isn't provided.	"Stop on Presence" logic requirement.

STPA Step 4: Identify Causal Loss Scenarios Exercise

2.8: Causal Loss Scenarios

For each UCA from Exercise 2.6, identify **at least one causal scenario** - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. **The causal factor:** What component failure, misunderstanding, or adverse condition triggers the UCA?
2. **Why it occurs:** What is the root cause or underlying reason?
3. **Connection to constraint:** Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

> **Hint:** Consider the following causal mechanisms (from the lecture):
>
> - **Incorrect feedback:** A sensor or perception model provides wrong information to the controller.
> - **Flawed internal model:** The controller misinterprets correct information (e.g., low confidence treated as "no detection").
> - **Timing delay:** Information arrives too late for the controller to respond safely.
> - **Human factor:** The human operator fails to intervene due to attention lapse, delayed reaction, or mode confusion.

Fill in the table below (add rows as needed):

UCA	Causal Scenario	Root Cause	Related Constraint
UCA-1	Detector reports "No Pedestrian".	Out-of-Distribution (OOD) case: pedestrian in wheelchair.	Pedestrian Recall Metric.
UCA-2	Braking not issued.	Overconfident error: low confidence treated as "No detection".	Calibration / ECE Requirement.
UCA-3	Braking issued too late.	High processing delay/latency in the perception pipeline.	Latency threshold requirement.
UCA-4	Operator fails to take over.	"Monitoring fatigue" during the end of a 4-hour shift.	Operator Shift Protocol (4h limit).
UCA-5	False Emergency Brake command.	Spurious correlation: model focuses on the sky/shadows, not objects.	Explainability (GradCAM) overlap.
UCA-6	Planner ignores traffic light.	Flawed internal model: assumes light is green because state isn't provided.	"Stop on Presence" logic requirement.

Connecting STPA to the Rest of the Course

Exercise 2.9: Mapping Constraints to Evidence *optional*

Review the safety constraints you derived in Exercise 2.7 and the causal scenarios from Exercise 2.8. For each **model-level** constraint, identify which future exercise sheet will provide evidence (refer to the Course Roadmap).

For each **system-level** constraint, briefly describe what kind of evidence or argument you would need - even though you cannot run an experiment for it in this course.

"This final step connects our theoretical safety rules to the actual coding experiments we will perform later in the semester to prove our system is safe."

Model-Level Safety Constraint	Relevant Exercise Sheet(s)	Type of Evidence Provided
High Detection Recall: The model must not miss pedestrians or vehicles in its path.	**Sheet 4: Testing I**	Precision, Recall, and F1 scores per model; Confusion matrices.
ODD Awareness: The system must identify when it is operating in conditions it was not trained for (e.g., rain or night).	**Sheet 9: Anomaly Detection**	OOD detection AUROC using MSP and energy scores for rain/fog/night scenes.
Reliable Confidence: The planner should not ignore a detection simply because the model is overconfident in an error.	**Sheet 7: Uncertainty**	Expected Calibration Error (ECE) and temperature scaling results.
Correct Feature Focus: The model must base its "Traffic Light" detection on the actual light, not spurious correlations like the sky.	**Sheet 8: Explainability**	GradCAM visualizations and overlap criterion evaluations.
Adversarial Robustness: The model must not be easily fooled by small, malicious changes to the input pixels.	**Sheet 6: Adversarial ML**	Recall metrics under FGSM attacks for various epsilon (\epsilon) values.

Evidence for System-Level Constraints

System-level constraints require design measures or human protocols that go beyond testing the machine learning models themselves. Since we cannot run physical experiments for these in this course, we must provide a structured **Safety Argument**.

1. Human Operator Fallback

- **Required Evidence:** You would need logs of operator reaction times and data from "eye-tracking" systems to prove the operator is actually monitoring the visual dashboard.
- **Argument:** "While the ML model may fail under extreme distribution shifts, the total system risk remains acceptable because a trained operator, restricted to 4-hour shifts to prevent fatigue, is positioned to override the actuators within 1.5 seconds of a perception failure".

2. Minimum Risk Maneuver (MRM)

- **Required Evidence:** Technical specifications for the rule-based planner showing that it triggers an emergency brake if **any** sensor feedback is lost or an ODD violation is detected.
- **Argument:** "The system is fail-safe because the rule-based planner does not require high-confidence detection to stop; if the Anomaly Detector (Sheet 9) flags an unknown environment, the planner is hard-coded to issue a maximum deceleration command to the actuators".

Summary of the "Safety Case" Connection

By the end of the course, we will combine all these pieces—the STPA constraints from today and the technical metrics from the coming weeks- into a final **Safety Case**.

This document will argue that the residual risk of operating the CARLA vehicle is bounded and acceptable for testing.

The system would have problem as there is no mechanism to detect the traffic light color just that the traffic light exists or not (Binary classifier) is there !!